PREPARED BY ADHYA RASTOGI

SEPTEMBER 18, 2025

SVEDKA Vodka

Demand and Pricing Analysis using SAS

CONSUMER AND MARKETING ANALYTICS

REGRESSION ANALYSIS

PRICE ELASTICITY

DEMAND MODELLING

DATA OVERVIEW AND SCHEMA

This analysis uses a panel dataset covering 31 vodka brands observed annually from 1995 to 2007, yielding 263 brand-year observations. The data combines cross-sectional brand characteristics with time-series variation in sales, pricing, and advertising spend, making it well-suited for demand modeling.

#	Variable	Type	Description
1	BrandName	Char	Vodka brand name
2	Brand_ID	Num	Unique brand identifier
3	Year	Num	Observation year (1995–2007)
4	TotalSales	Num	Annual unit sales volume
5	Domestic	Num	1 = Domestic brand, 0 = Imported
6	DollarSales	Num	Total revenue in USD
7	PricePerUnit	Num	Price per unit (USD)
8	Mag	Num	Magazine advertising spend
9	News	Num	Newspaper advertising spend
10	Outdoor	Num	Outdoor advertising spend
11	Broad	Num	Broadcast (TV/Radio) advertising spend
12	Print	Num	Print advertising spend
13	Tier	Num	Brand tier: 1=Premium, 2=Mid, 3=Mass-market
14	TotalMinusSales	Num	Category volume excluding focal brand
15	Firstintro	Num	1 = Brand newly introduced that year
16	Marketshare	Num	Brand's share of category volume
17	YearID	Num	Sequential year index (1–15)

MARKET LANDSCAPE – DESCRIPTIVE ANALYSIS

Exploratory analysis reveals a highly fragmented and competitive market with meaningful variation across pricing tiers, brand origin, and advertising strategy.

Sales and Pricing Distribution

Variable	Mean	Std Dev	Min	Max
TotalSales (units)	1,400	1,598	42	9,015
PricePerUnit ($)	$83.71	$57.13	$27.29	$250.26
DollarSales ($)	$113,913	$166,366	$0	$786,721
Marketshare	0.05	0.05	0.00	0.27

Advertising Spend by Channel

Channel	Mean Spend	Std Dev	Max Observed
Broadcast (TV/Radio)	$276	$1,044	$7,827
Magazine	$3,280	$6,691	$33,971
Print	$3,474	$6,878	$34,505
Outdoor	$162	$467	$3,255
Newspaper	$194	$550	$3,525

Brand Composition

Segment	Count	Share	Notes
Tier 3 — Mass-market	137	52.1%	Highest volume, most price-sensitive
Tier 2 — Mid-range	65	24.7%	Balanced price/quality positioning
Tier 1 — Premium	61	23.2%	High price, lower volume
Domestic brands	161	61.2%	Majority of portfolio
Imported brands	102	38.8%	Higher avg. price point
New brand introductions	4	1.5%	Mature, stable category

Key Insight: SVEDKA operates in a mature and fragmented vodka category. Most competing brands are in the mass-market tier, advertising spend varies dramatically (some brands investing nearly nothing, others exceeding $7,800 in broadcast alone), and only 4 new brands entered over 13 years, indicating a stable, loyalty-driven competitive landscape.

REGRESSION ANALYSIS I – PRICE AS A DEMAND DRIVER

The first model tests the direct relationship between unit price and sales volume, establishing a baseline for understanding price sensitivity in the vodka category.

Model Specification

$$TotalSales = \beta 0 + \beta 1 (PricePerUnit) + \varepsilon$$

Results

Parameter	Estimate	Std Error	t-Value	p-Value
Intercept	1,485.84	175.25	8.48	< .0001
PricePerUnit	–1.02	1.73	–0.59	0.556

Model Fit Metric	Value	Interpretation
R^2	0.0013	Price explains 0.13% of sales variation
Adj. R^2	–0.003	No meaningful predictive power
F-statistic	0.35	Model not significant overall
Root MSE	1,599.9	Large residual error

Price alone is a poor predictor of vodka sales. The negative coefficient (-1.02) is directionally consistent with demand theory, but it is statistically insignificant ($p = 0.556$) and explains less than 1% of sales variance. This is likely due to pooling brands across very different price tiers, premium brands sell at $250 but with high loyalty, while budget brands at $27 compete entirely on volume. Raw price comparison across tiers obscures true price effects.

REGRESSION ANALYSIS II – PRICING AND ADVERTISING

Recognizing that advertising is a central competitive lever for SVEDKA, the model was expanded to include all five advertising channel variables. Best-subset selection (R^2 criterion) identified the optimal specification.

Final Model Specification

TotalSales = β 0 + β 1 (PricePerUnit) + β 2 (Broadcast) + β 3 (Outdoor) + β 4 (Magazine) + β 5 (News) + ε

Parameter Estimates

Variable	Coefficient	Std Error	t-Value	p-Value	Significance
Intercept	1,582.55	96.85	16.34	< .0001	✔
PricePerUnit	–10.48	1.10	–9.54	< .0001	✔
Broadcast	+0.92	0.055	16.77	< .0001	✔ (strongest)
Outdoor	+0.61	0.152	4.04	< .0001	✔
Magazine	+0.097	0.011	8.84	< .0001	✔
News	+0.13	0.108	1.18	0.239	✘ (not sig.)

Model Fit Metric	Simple Model	Full Model	Improvement
R^2	0.0013	0.708	+70.7 percentage points
Adj. R^2	–0.003	0.702	Robust across variables
Root MSE	1,599.9	872.1	45% reduction in prediction error

Adding advertising channels increased explained variance from 0.13% to 70.8%, a dramatic improvement. Price became highly significant ($p < .0001$) once advertising was controlled for, confirming that omitting advertising created omitted variable bias in the simple model. Broadcast

advertising had by far the strongest marginal effect on sales volume, followed by Outdoor and Magazine channels. Newspaper advertising showed no statistically meaningful impact.

PRICE ELASTICITY OF DEMAND

Price elasticity quantifies how responsive sales volume is to a percentage change in price. It is computed at the sample means using the regression coefficient from the simple price model, isolating the pure price sensitivity of the average brand in the dataset.

Elasticity Formula & Calculation

Using the regression coefficient of PricePerUnit and the sample means of price and sales; I computed the price elasticity of demand to determine how responsive sales are to price changes. The formula for elasticity is:

EP = βPrice X Mean Price(P)/Mean Sales(Q)

βPrice= coefficient of price

Where:

βPrice = −1.02042(regression coefficient)

P=83.71 (mean price per unit)

Q=1400.41 (mean total sales)

Calculation: EP = -1.02042 X 83.71/1400.41 = -0.060997

Elasticity Value	Category	Meaning
−0.061	Highly Inelastic	A 1% price increase reduces sales by only 0.06%
|E| < 1	Inelastic demand	Consumers are insensitive to price changes
|E| ≈ 0	Near-zero effect	Revenue increases with modest price hikes

Vodka demand is highly price inelastic (EP = −0.061). Even a 10% price increase would reduce sales by less than 1% on average, suggesting that vodka is a habitual, brand-loyal category where consumers prioritize preference over price. This gives brands meaningful pricing power, particularly in the premium and mid-range tiers.

STRATEGIC RECOMMENDATIONS

The regression results and elasticity estimates translate directly into actionable strategy across three domains: pricing, advertising allocation, and market segmentation.

Pricing Strategy: With an elasticity of -0.061, demand is nearly price-insensitive. Moderate price increases (5-10%) are unlikely to erode volume meaningfully and would directly improve margin. Caution is warranted at extreme price points where elasticity may shift, but within the observed price range this represents a low-risk revenue lever

Advertising Allocation: Broadcast (TV/Radio) advertising delivers the strongest marginal return on sales; brands should prioritize this channel. Magazine and Outdoor also contribute positively. Newspaper advertising showed no statistically significant effect (p = 0.239) and represents budget that could be reallocated to higher-ROI channels without volume risk.

Tier Segmentation: Mass-market (Tier 3) brands dominate by count but likely face sharper price sensitivity within their segment. Premium (Tier 1) brands can sustain higher prices through advertising investment that reinforces brand equity. A tiered pricing strategy, stable mass-market pricing with selective premium price increases, optimizes portfolio revenue.

Bottom Line: The model explains 70.8% of sales variation, providing a reliable quantitative foundation for decision-making. The optimal strategy combines modest price increases with a reallocation of advertising spend toward broadcast and away from print/news, expected to simultaneously grow revenue and improve marketing efficiency in a mature, competitive category.

A: APPENDIX – ANNOTATED SAS CODE

All analysis was conducted in SAS. The code below is organized by analytical step and follows the same structure as the report sections above.

Data Import and Validation

```
/* -- Import via DATA step -- */

libname vd 'path/to/data';


DATA vd.vodka;

INFILE 'svedka.csv' delimiter=',' FIRSTOBS=2;

INPUT BrandName $ Brand_ID Year TotalSales Domestic DollarSales

PricePerUnit Mag News Outdoor Broad Print Tier

TotalMinusSales Firstintro Marketshare YearID;

RUN;


/* -- Alternative: PROC IMPORT -- */

PROC IMPORT OUT=vd.vodka DATAFILE='svedka.csv' DBMS=CSV REPLACE;

GETNAMES=YES; DATAROW=2;

RUN;


PROC CONTENTS DATA=vd.vodka; RUN;

PROC PRINT DATA=vd.vodka (obs=10); RUN;
```

Exploratory Data Analysis

```
/* -- Summary statistics -- */

PROC MEANS DATA=vd.vodka maxdec=2; RUN;


/* -- Categorical distributions -- */

PROC FREQ DATA=vd.vodka;

TABLES Tier Domestic FirstIntro Year;

RUN;
```

```
/* -- Cross-tab by tier and year -- */

PROC TABULATE DATA=vd.vodka;

CLASS Tier Year;

VAR TotalSales PricePerUnit;

TABLE Tier, (TotalSales PricePerUnit)*(mean std min max);

RUN;
```

Regression – Price Only

```
PROC REG DATA=vd.vodka PLOTS=none OUTEST=linreg_vodka;

MODEL TotalSales = PricePerUnit;

RUN; QUIT;


PROC PRINT DATA=linreg_vodka; RUN;
```

Regression – Price & Advertising (Best Subset)

```
PROC REG DATA=vd.vodka plots=none OUTEST=linreg_vd;

MODEL TotalSales = PricePerUnit Outdoor Broad Mag News

/ selection=rsquare b best=1;

RUN; QUIT;


PROC PRINT DATA=linreg_vd; RUN;
```

Price Elasticity Estimation

```
/* -- Extract sample means -- */

PROC MEANS DATA=vd.vodka mean;

VAR TotalSales PricePerUnit Mag News Outdoor Broad;

OUTPUT OUT=mean_ads MEAN= /autoname;

RUN;
```

```
/* -- Merge regression output with means and compute elasticity -- */

DATA elasticity_vodka_price;

MERGE linreg_vodka(drop=_TYPE_) mean_ads(drop=_TYPE_);

E_P_PricePerUnit = PricePerUnit * (PricePerUnit_Mean / TotalSales_Mean);

OUTPUT;

RUN;


PROC PRINT DATA=elasticity_vodka_price; RUN;
```

B: APPENDIX – OUTPUT FROM THE ABOVE CODE

Data Summary

Alphabetic List of Variables and Attributes					
#	Variable	Type	Len	Format	Informat
1	BrandName	Char	9	$9.	$9.
2	Brand_ID	Num	8	BEST12.	BEST32.
11	Broad	Num	8	BEST12.	BEST32.
6	DollarSales	Num	8	BEST12.	BEST32.
5	Domestic	Num	8	BEST12.	BEST32.
15	Firstintro	Num	8	BEST12.	BEST32.
8	Mag	Num	8	BEST12.	BEST32.
16	Marketshare	Num	8	BEST12.	BEST32.
9	News	Num	8	BEST12.	BEST32.
10	Outdoor	Num	8	BEST12.	BEST32.
7	PricePerUnit	Num	8	BEST12.	BEST32.
12	Print	Num	8	BEST12.	BEST32.
13	Tier	Num	8	BEST12.	BEST32.
14	TotalMinusSales	Num	8	BEST12.	BEST32.
4	TotalSales	Num	8	BEST12.	BEST32.
3	Year	Num	8	BEST12.	BEST32.
17	YearID	Num	8	BEST12.	BEST32.

Summary Statistics and Initial Inference

The SAS System

The MEANS Procedure

Variable	N	Mean	Std Dev	Minimum	Maximum
Brand_ID	263	12.60	7.65	1.00	31.00
Year	263	2001.70	3.64	1995.00	2007.00
TotalSales	263	1400.41	1597.96	42.00	9015.00
Domestic	263	0.61	0.49	0.00	1.00
DollarSales	263	113912.88	166365.75	4690.00	786721.00
PricePerUnit	263	83.71	57.13	27.29	250.26
Mag	263	3280.29	6690.51	1.00	33971.30
News	263	193.91	549.92	0.30	3524.90
Outdoor	263	162.04	467.14	1.00	3255.40
Broad	263	275.92	1043.66	0.60	7827.10
Print	263	3474.20	6878.26	2.00	34504.70
Tier	263	2.29	0.82	1.00	3.00
TotalMinusSales	263	62653.85	1599.77	55209.00	64163.00
Firstintro	263	0.02	0.12	0.00	1.00
Marketshare	263	0.05	0.05	0.00	0.27
YearID	263	9.70	3.64	3.00	15.00

The SAS System

The FREQ Procedure

Tier	Frequency	Percent	Cumulative Frequency	Cumulative Percent
1	61	23.19	61	23.19
2	65	24.71	126	47.91
3	137	52.09	263	100.00

Domestic	Frequency	Percent	Cumulative Frequency	Cumulative Percent
0	102	38.78	102	38.78
1	161	61.22	263	100.00

Firstintro	Frequency	Percent	Cumulative Frequency	Cumulative Percent
0	259	98.48	259	98.48
1	4	1.52	263	100.00

Simple Linear Regression – Sales on Price

The SAS System

The REG Procedure
Model: MODEL1
Dependent Variable: TotalSales

Number of Observations Read	263
Number of Observations Used	263

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	890287	890287	0.35	0.5559
Error	261	668119472	2559845		
Corrected Total	262	669009760			

Root MSE	1599.95147	R-Square	0.0013
Dependent Mean	1400.41445	Adj R-Sq	-0.0025
Coeff Var	114.24843		

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	1485.83515	175.25254	8.48	<.0001
PricePerUnit	1	-1.02042	1.73030	-0.59	0.5559

The SAS System

Obs	_MODEL_	_TYPE_	_DEPVAR_	_RMSE_	Intercept	PricePerUnit	TotalSales
1	MODEL1	PARMS	TotalSales	1599.95	1485.84	-1.02042	-1

Regression with Price and Advertising

The SAS System

The REG Procedure
Model: MODEL1
Dependent Variable: TotalSales

Number of Observations Read	263
Number of Observations Used	263

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	5	473557861	94711572	124.54	<.0001
Error	257	195451899	760513		
Corrected Total	262	669009760			

Root MSE	872.07409	R-Square	0.7078
Dependent Mean	1400.41445	Adj R-Sq	0.7022
Coeff Var	62.27257		

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	1582.55091	96.85015	16.34	<.0001
PricePerUnit	1	-10.47726	1.09821	-9.54	<.0001
Outdoor	1	0.61292	0.15163	4.04	<.0001
Broad	1	0.92121	0.05492	16.77	<.0001
Mag	1	0.09654	0.01092	8.84	<.0001
News	1	0.12754	0.10812	1.18	0.2392

Price Elasticity of Demand

The SAS System

The MEANS Procedure

Variable	Mean
TotalSales	1400.41
PricePerUnit	83.7111141
Mag	3280.29
News	193.9098859
Outdoor	162.0410646
Broad	275.9220532

The SAS System

Obs	_TYPE_	_FREQ_	TotalSales_Mean	PricePerUnit_Mean	Mag_Mean	News_Mean	Outdoor_Mean	Broad_Mean
1	0	263	1400.4144487	83.711114144	3280.2882129	193.90988593	162.04106464	275.92205323

The SAS System

Obs	_MODEL_	_DEPVAR_	_RMSE_	Intercept	PricePerUnit	TotalSales	_FREQ_	TotalSales_Mean	PricePerUnit_Mean	Mag_Mean	News_Mean	Outdoor_Mean	Broad_Mean	E_P_PricePerUnit
1	MODEL1	TotalSales	1599.95	1485.84	-1.02042	-1	263	1400.4144487	83.711114144	3280.2882129	193.90988593	162.04106464	275.92205323	-0.060997